

# Clinical Effectiveness

## METIMMOX-1 Economic Evaluation

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### 1 Overview

This report presents clinical effectiveness results from the METIMMOX-1 trial, evaluating biomarker-guided treatment strategies for selecting immunotherapy candidates in metastatic MSS/pMMR colorectal cancer.

The analysis uses **Firth-corrected Cox proportional hazards regression** with profile likelihood confidence intervals to assess treatment effect heterogeneity across three candidate biomarkers:

- **CRP** (C-reactive protein): Low CRP (<5 mg/L) at baseline
- **TLR** (Tumor Lesion Reduction): Early response (10% reduction at first CT scan)
- **TMB/BRAF**: High tumor mutational burden (9 mut/MB) or BRAF mutation

We compare three model structures to assess the robustness of treatment-biomarker interactions:

- **Model A**: All-in-one model with all biomarkers and interactions
- **Model B**: Focused models with all biomarkers but only one interaction per model
- **Model C**: Single-biomarker models with one biomarker and its interaction

## 2 Methodological Notes

### 2.1 Software Packages

This analysis uses the following R packages:

- **coxphf** version 1.13.4: Firth-corrected Cox regression with profile likelihood CIs
- **survival** version 3.7.0: Cox regression and Schoenfeld residual diagnostics
- **glmnet** version 4.1.8: Ridge regression sensitivity analysis
- **vcd** version 1.4.13: Cramér's V for biomarker correlation assessment

### 2.2 Firth-Corrected Cox Regression

We use Firth's penalized maximum likelihood method for Cox regression, which addresses:

1. **Monotone likelihood**: When a covariate perfectly separates events from non-events
2. **Small sample bias**: Reduces bias in coefficient estimates with sparse data
3. **Unstable confidence intervals**: Profile likelihood CIs are more reliable than Wald-based CIs in small samples

Statistical inference is based on **penalized likelihood ratio tests (PLRT)**, which are more appropriate than Wald tests for testing treatment-biomarker interactions in this setting.

### 2.3 Ridge Regression Sensitivity Analysis

We use Ridge regression as a sensitivity analysis to assess the stability of our interaction estimates. Ridge regression (also called L2-penalized regression or Tikhonov regularization) belongs to the family of regularized or penalized regression methods. In survival analysis, this is implemented as a penalized Cox model.

**How Ridge regression works:**

- Ridge adds a penalty term that discourages large coefficients. The bigger a coefficient tries to get, the more the model resists, pulling estimates toward zero (HR toward 1.0).
- Unlike LASSO ( $\alpha = 1$ ), which sets some coefficients exactly to zero and performs variable selection, Ridge ( $\alpha = 0$ ) shrinks all coefficients but retains all predictors.
- The degree of shrinkage is controlled by the tuning parameter  $\lambda$ , which is selected via **cross-validation (CV)**: the data are repeatedly split into training and validation sets, and the  $\lambda$  that produces the best predictive performance is chosen.

**Why not LASSO or Elastic Net?**

We determined that LASSO and elastic net are **not appropriate** for this analysis:

1. **Correlated biomarkers:** LASSO arbitrarily selects one of correlated predictors while setting others to zero, which does not reflect the clinical question of comparing ALL biomarker strategies.
2. **Hierarchy principle violation:** LASSO may select interaction terms (e.g., `crp:Rx`) while setting the main effect (`crp`) to zero, producing models that are difficult to interpret clinically.
3. **Clinical objective:** Our goal is to estimate treatment effect heterogeneity across ALL biomarkers simultaneously, not to select a “winning” biomarker.

Elastic net ( $\alpha = 0.1\text{--}0.3$ ) represents a compromise but does not fully address these concerns.

### Interpreting Ridge results:

Ridge regression is not replacing the Firth estimates—it provides a reality check. When Ridge substantially shrinks an interaction coefficient toward zero while Firth estimates a large effect, this indicates the effect is too unstable to trust for practical use. The CV-selected lambda reflects how much regularization is needed for reasonable out-of-sample prediction given the sample size and collinearity structure.

## 2.4 Multiple Testing Correction

We apply the **Benjamini-Hochberg (BH) procedure** to control the false discovery rate (FDR) on the primary Firth analysis:

- 6 interaction tests: 3 biomarkers  $\times$  2 outcomes (OS, PFS)
- Both raw PLRT p-values and BH-adjusted q-values are reported
- The Ridge sensitivity analysis does not require separate multiplicity correction

## 3 Data Preparation

### 3.1 Sample Characteristics

Table 1: Sample Characteristics

| Characteristic                                | N (%)      |
|-----------------------------------------------|------------|
| Total patients (complete cases)               | 65         |
| Control arm (FLOX alone)                      | 29 (44.6%) |
| Experimental arm (alternating FLOX/nivolumab) | 36 (55.4%) |
| Deaths (OS events)                            | 56 (86.2%) |
| Progressions (PFS events)                     | 48 (73.8%) |
| CRP-positive (low CRP)                        | 22 (33.8%) |
| TLR-positive (early response)                 | 41 (63.1%) |
| TMB/BRAF-positive                             | 29 (44.6%) |

### 3.2 Biomarker Correlations

**Note:** Cramér’s V interpretation:  $<0.1$  = Negligible,  $0.1\text{--}0.3$  = Weak,  $>0.3$  = Moderate/Strong. The observed correlations are weak to negligible. While modest, any correlation among candidate biomarkers supports our decision not to use LASSO, which would arbitrarily select among correlated predictors.

Table 2: Biomarker Correlations (Cramér’s V)

| Biomarker Pair  | Cramér’s V | Interpretation |
|-----------------|------------|----------------|
| CRP vs TLR      | 0.278      | Weak           |
| CRP vs TMB/BRAF | 0.077      | Negligible     |
| TLR vs TMB/BRAF | 0.019      | Negligible     |

## 4 Proportional Hazards Assumption

### 4.1 Schoenfeld Residual Tests

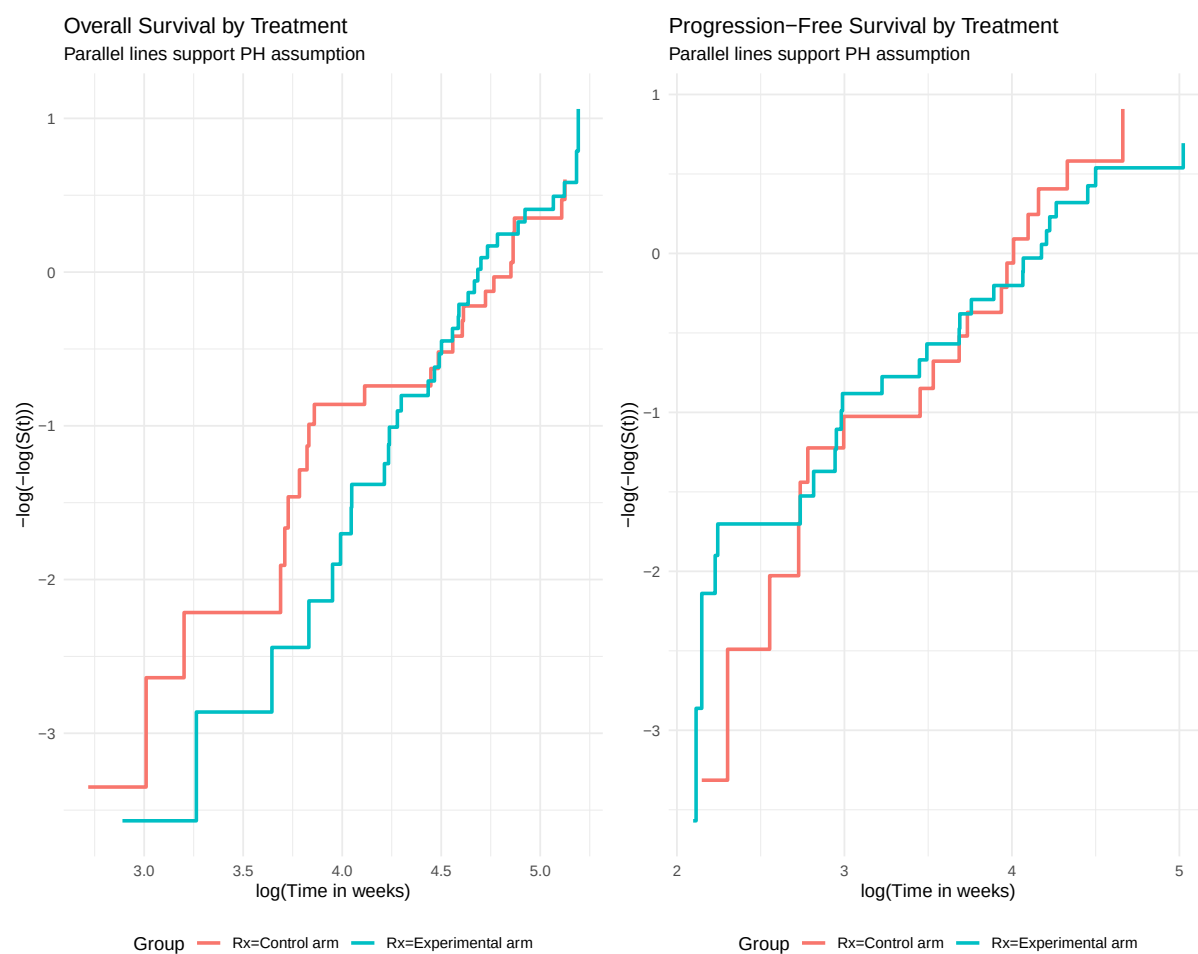
The proportional hazards assumption was tested using scaled Schoenfeld residuals. A significant p-value indicates potential violation of the PH assumption for that covariate.

Table 3: Schoenfeld Residual Test for Proportional Hazards Assumption

| Covariate       | $\chi^2$ (OS) | p-value      | $\chi^2$ (PFS) | p-value      |
|-----------------|---------------|--------------|----------------|--------------|
| Age             | 2.84          | 0.092        | 0.42           | 0.515        |
| sex             | 0.55          | 0.460        | 0.13           | 0.715        |
| Rx              | 1.05          | 0.306        | 0.31           | 0.579        |
| crp_num         | 0.37          | 0.543        | 1.36           | 0.243        |
| tlr_num         | 0.19          | 0.665        | 1.00           | 0.318        |
| tmb_braf_num    | 0.57          | 0.449        | 8.43           | 0.004        |
| Rx:crp_num      | 1.65          | 0.199        | 0.35           | 0.554        |
| Rx:tlr_num      | 1.98          | 0.159        | 1.60           | 0.206        |
| Rx:tmb_braf_num | 0.18          | 0.676        | 6.78           | 0.009        |
| <b>GLOBAL</b>   | <b>7.38</b>   | <b>0.598</b> | <b>24.23</b>   | <b>0.004</b> |

**Note:** GLOBAL is the omnibus test for all covariates. Individual covariate tests help identify specific PH violations.

## 4.2 Log-Log Survival Plots



**Interpretation:** Approximately parallel lines on the log-log plot support the proportional hazards assumption. Crossing or converging lines suggest time-varying effects.

## 5 Primary Analysis: Firth-Corrected Cox Models

### 5.1 Model Type A: All-in-One Model

This model includes all three biomarkers as main effects and all three treatment-biomarker interactions simultaneously.

**Formula:** `Surv(time, event) ~ Age + sex + Rx + crp + tlr + tmb_braf + crp:Rx + tlr:Rx + tmb_braf:Rx`

#### 5.1.1 Overall Survival (Model A)

Table 4: Model A Overall Survival - Full Coefficient Table

| Term                                      | HR (95% CI)      | PLRT p-value | BH-adjusted p |
|-------------------------------------------|------------------|--------------|---------------|
| <b>Treatment × Biomarker Interactions</b> |                  |              |               |
| RxExperimental arm:crp_num                | 0.71 (0.17-3.56) | 1.000        | 1.000         |
| RxExperimental arm:tlr_num                | 2.63 (0.75-8.74) | 1.000        | 1.000         |
| RxExperimental arm:tmb_braf_num           | 0.92 (0.29-2.91) | 1.000        | 1.000         |
| <b>Main Effects and Covariates</b>        |                  |              |               |
| Age                                       | 1.01 (0.98-1.04) | —            | —             |
| sex1                                      | 1.48 (0.82-2.68) | —            | —             |
| RxExperimental arm                        | 0.62 (0.23-1.77) | —            | —             |
| crp_num                                   | 0.70 (0.18-2.06) | —            | —             |
| tlr_num                                   | 0.20 (0.08-0.55) | —            | —             |
| tmb_braf_num                              | 0.93 (0.40-2.12) | —            | —             |

#### 5.1.2 Progression-Free Survival (Model A)

Table 5: Model A Progression-Free Survival - Full Coefficient Table

| Term                                      | HR (95% CI)      | PLRT p-value | BH-adjusted p |
|-------------------------------------------|------------------|--------------|---------------|
| <b>Treatment × Biomarker Interactions</b> |                  |              |               |
| RxExperimental arm:crp_num                | 0.22 (0.04-1.59) | 1.000        | 1.000         |
| RxExperimental arm:tlr_num                | 0.87 (0.22-3.22) | 1.000        | 1.000         |
| RxExperimental arm:tmb_braf_num           | 0.48 (0.12-1.93) | 1.000        | 1.000         |
| <b>Main Effects and Covariates</b>        |                  |              |               |
| Age                                       | 1.00 (0.97-1.03) | —            | —             |
| sex1                                      | 1.26 (0.64-2.52) | —            | —             |
| RxExperimental arm                        | 1.88 (0.61-6.61) | —            | —             |
| crp_num                                   | 1.75 (0.31-7.00) | —            | —             |
| tlr_num                                   | 0.14 (0.05-0.45) | —            | —             |
| tmb_braf_num                              | 0.80 (0.30-2.09) | —            | —             |

## 5.2 Model B: Focused Models

These models include all three biomarkers as main effects but only ONE treatment-biomarker interaction per model. This approach tests each interaction while controlling for all biomarkers.

Table 6: Model B: Focused Models - Treatment  $\times$  Biomarker Interaction Summary

| Interaction Term            | Overall Survival |        | Progression-Free Survival |        |
|-----------------------------|------------------|--------|---------------------------|--------|
|                             | HR (95% CI)      | PLRT p | HR (95% CI)               | PLRT p |
| CRP $\times$ Treatment      | 0.97 (0.26-4.54) | 1.000  | 0.19 (0.03-1.34)          | 1.000  |
| TLR $\times$ Treatment      | 2.47 (0.75-7.75) | 1.000  | 0.75 (0.20-2.66)          | 1.000  |
| TMB/BRAF $\times$ Treatment | 0.89 (0.28-2.79) | 1.000  | 0.45 (0.11-1.78)          | 1.000  |

**Note:** Each row represents a separate model with all biomarkers as main effects but only the specified interaction.

### 5.2.1 Full Coefficient Tables

Table 7: Model B CRP-Focused: Overall Survival - Full Coefficients

| Term                       | HR (95% CI)      | p-value |
|----------------------------|------------------|---------|
| RxExperimental arm:crp_num | 0.97 (0.26-4.54) | 0.970   |
| Age                        | 1.01 (0.98-1.04) | 0.553   |
| sex1                       | 1.53 (0.86-2.77) | 0.148   |
| RxExperimental arm         | 1.02 (0.48-2.15) | 0.961   |
| crp_num                    | 0.61 (0.16-1.73) | 0.375   |
| tlr_num                    | 0.37 (0.19-0.71) | 0.003   |
| tmb_braf_num               | 0.87 (0.48-1.56) | 0.651   |

Table 8: Model B CRP-Focused: Progression-Free Survival - Full Coefficients

| Term                       | HR (95% CI)      | p-value |
|----------------------------|------------------|---------|
| RxExperimental arm:crp_num | 0.19 (0.03-1.34) | 0.090   |
| Age                        | 1.00 (0.96-1.03) | 0.886   |
| sex1                       | 1.22 (0.63-2.39) | 0.558   |
| RxExperimental arm         | 1.42 (0.67-3.00) | 0.362   |
| crp_num                    | 1.58 (0.29-5.89) | 0.549   |
| tlr_num                    | 0.13 (0.06-0.29) | <0.001  |
| tmb_braf_num               | 0.56 (0.27-1.14) | 0.111   |

Table 9: Model B TLR-Focused: Overall Survival - Full Coefficients

| Term                       | HR (95% CI)      | p-value |
|----------------------------|------------------|---------|
| RxExperimental arm:tlr_num | 2.47 (0.75-7.75) | 0.135   |
| Age                        | 1.01 (0.98-1.04) | 0.626   |
| sex1                       | 1.47 (0.82-2.67) | 0.201   |
| RxExperimental arm         | 0.59 (0.24-1.55) | —       |
| crp_num                    | 0.53 (0.25-1.05) | 0.071   |
| tlr_num                    | 0.21 (0.08-0.55) | 0.002   |
| tmb_braf_num               | 0.86 (0.48-1.52) | 0.609   |

Table 10: Model B TLR-Focused: Progression-Free Survival - Full Coefficients

| Term                       | HR (95% CI)      | p-value |
|----------------------------|------------------|---------|
| RxExperimental arm:tlr_num | 0.75 (0.20-2.66) | 0.660   |
| Age                        | 1.00 (0.96-1.03) | 0.787   |
| sex1                       | 1.22 (0.62-2.45) | 0.570   |
| RxExperimental arm         | 1.35 (0.48-4.22) | 0.576   |
| crp_num                    | 0.43 (0.18-0.96) | 0.039   |
| tlr_num                    | 0.17 (0.06-0.52) | 0.003   |
| tmb_braf_num               | 0.50 (0.24-0.99) | 0.046   |

Table 11: Model B TMB/BRAF-Focused: Overall Survival - Full Coefficients

| Term                            | HR (95% CI)      | p-value |
|---------------------------------|------------------|---------|
| RxExperimental arm:tmb_braf_num | 0.89 (0.28-2.79) | 0.844   |
| Age                             | 1.01 (0.98-1.04) | 0.575   |
| sex1                            | 1.54 (0.86-2.78) | 0.143   |
| RxExperimental arm              | 1.07 (0.50-2.33) | 0.860   |
| crp_num                         | 0.59 (0.28-1.19) | 0.141   |
| tlr_num                         | 0.37 (0.19-0.71) | 0.003   |
| tmb_braf_num                    | 0.94 (0.40-2.11) | 0.879   |

Table 12: Model B TMB/BRAF-Focused: Progression-Free Survival - Full Coefficients

| Term                            | HR (95% CI)      | p-value |
|---------------------------------|------------------|---------|
| RxExperimental arm:tmb_braf_num | 0.45 (0.11-1.78) | –       |
| Age                             | 0.99 (0.96-1.03) | 0.715   |
| sex1                            | 1.18 (0.61-2.32) | 0.631   |
| RxExperimental arm              | 1.41 (0.63-3.26) | –       |
| crp_num                         | 0.52 (0.20-1.27) | –       |
| tlr_num                         | 0.14 (0.07-0.30) | <0.001  |
| tmb_braf_num                    | 0.71 (0.27-1.76) | 0.459   |

### 5.3 Model C: Single-Biomarker Models

These models include only ONE biomarker and its treatment interaction. This is the simplest specification but does not control for other biomarkers.

Table 13: Model C: Single-Biomarker Models - Treatment  $\times$  Biomarker Interaction Summary

| Interaction Term            | Overall Survival |        | Progression-Free Survival |        |
|-----------------------------|------------------|--------|---------------------------|--------|
|                             | HR (95% CI)      | PLRT p | HR (95% CI)               | PLRT p |
| CRP $\times$ Treatment      | 0.72 (0.20-3.23) | 1.000  | 0.21 (0.05-1.31)          | 1.000  |
| TLR $\times$ Treatment      | 1.98 (0.61-6.06) | 1.000  | 0.53 (0.14-1.86)          | 1.000  |
| TMB/BRAF $\times$ Treatment | 0.59 (0.20-1.72) | 1.000  | 0.36 (0.10-1.25)          | 1.000  |

**Note:** Each row represents a separate model with only that biomarker and its interaction (no other biomarkers included).

### 5.3.1 Full Coefficient Tables

Table 14: Model C CRP-Only: Overall Survival - Full Coefficients

| Term                       | HR (95% CI)      | p-value |
|----------------------------|------------------|---------|
| RxExperimental arm:crp_num | 0.72 (0.20-3.23) | 0.644   |
| Age                        | 1.00 (0.97-1.03) | 0.995   |
| sex1                       | 1.60 (0.92-2.81) | 0.095   |
| RxExperimental arm         | 1.71 (0.89-3.25) | 0.104   |
| crp_num                    | 0.53 (0.14-1.48) | 0.241   |

Table 15: Model C CRP-Only: Progression-Free Survival - Full Coefficients

| Term                       | HR (95% CI)      | p-value |
|----------------------------|------------------|---------|
| RxExperimental arm:crp_num | 0.21 (0.05-1.31) | 0.089   |
| Age                        | 0.98 (0.95-1.01) | 0.288   |
| sex1                       | 1.27 (0.69-2.38) | 0.451   |
| RxExperimental arm         | 1.90 (0.96-3.78) | 0.065   |
| crp_num                    | 1.10 (0.22-3.62) | 0.887   |

Table 16: Model C TLR-Only: Overall Survival - Full Coefficients

| Term                       | HR (95% CI)      | p-value |
|----------------------------|------------------|---------|
| RxExperimental arm:tlr_num | 1.98 (0.61-6.06) | 0.249   |
| Age                        | 1.01 (0.98-1.04) | 0.713   |
| sex1                       | 1.59 (0.90-2.83) | 0.111   |
| RxExperimental arm         | 0.52 (0.21-1.37) | —       |
| tlr_num                    | 0.19 (0.08-0.51) | 0.002   |

Table 17: Model C TLR-Only: Progression-Free Survival - Full Coefficients

| Term                       | HR (95% CI)      | p-value |
|----------------------------|------------------|---------|
| RxExperimental arm:tlr_num | 0.53 (0.14-1.86) | 0.328   |
| Age                        | 0.99 (0.96-1.03) | 0.740   |
| sex1                       | 1.67 (0.87-3.26) | 0.123   |
| RxExperimental arm         | 1.10 (0.41-3.25) | 0.856   |
| tlr_num                    | 0.23 (0.08-0.66) | 0.008   |

Table 18: Model C TMB/BRAF-Only: Overall Survival - Full Coefficients

| Term                            | HR (95% CI)      | p-value |
|---------------------------------|------------------|---------|
| RxExperimental arm:tmb_braf_num | 0.59 (0.20-1.72) | 0.328   |
| Age                             | 0.99 (0.97-1.02) | 0.696   |
| sex1                            | 1.52 (0.87-2.70) | 0.141   |
| RxExperimental arm              | 1.44 (0.73-2.92) | 0.295   |
| tmb_braf_num                    | 1.03 (0.44-2.31) | 0.945   |

Table 19: Model C TMB/BRAF-Only: Progression-Free Survival - Full Coefficients

| Term                            | HR (95% CI)      | p-value |
|---------------------------------|------------------|---------|
| RxExperimental arm:tmb_braf_num | 0.36 (0.10-1.25) | 0.107   |
| Age                             | 0.98 (0.95-1.01) | 0.244   |
| sex1                            | 1.19 (0.64-2.25) | 0.588   |
| RxExperimental arm              | 1.40 (0.67-3.05) | 0.372   |
| tmb_braf_num                    | 0.77 (0.29-1.94) | 0.585   |

## 6 Model Comparison

### 6.1 Summary: Interaction HRs Across Model Types

Table 20: Overall Survival: Treatment  $\times$  Biomarker Interactions Across Model Types

| Interaction          | Model A          | Model B          | Model C          |
|----------------------|------------------|------------------|------------------|
| CRP $\times$ Rx      | 0.71 (0.17-3.56) | 0.97 (0.26-4.54) | 0.72 (0.20-3.23) |
| TLR $\times$ Rx      | 2.63 (0.75-8.74) | 2.47 (0.75-7.75) | 1.98 (0.61-6.06) |
| TMB/BRAF $\times$ Rx | 0.92 (0.29-2.91) | 0.89 (0.28-2.79) | 0.59 (0.20-1.72) |

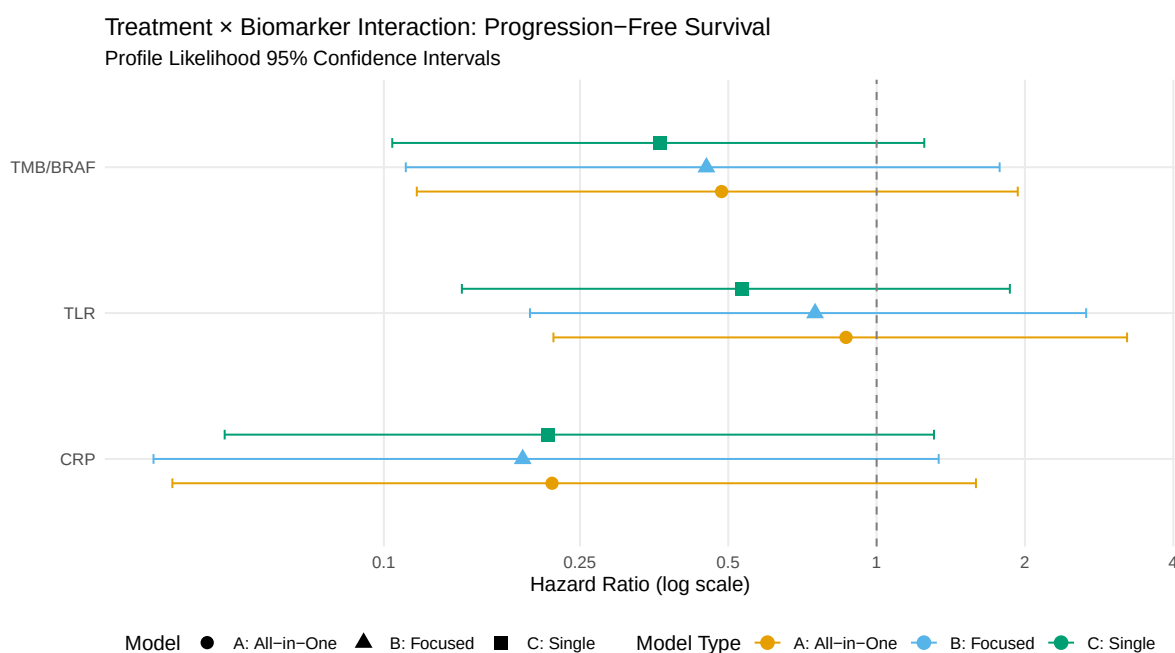
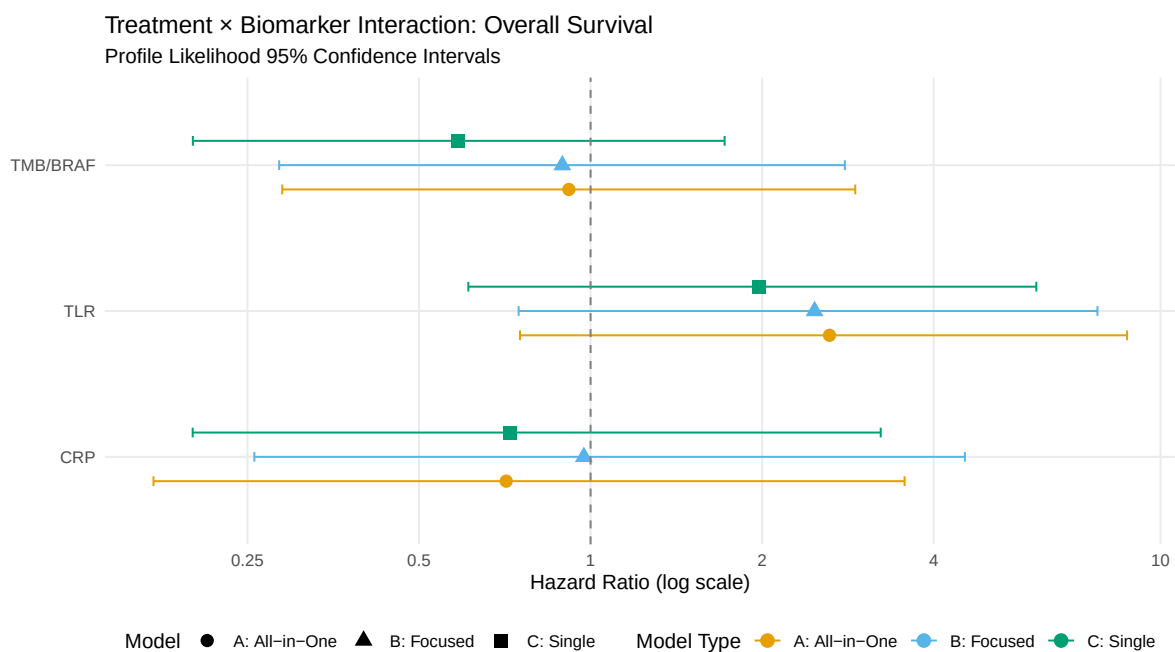
*Note:* Model A: All-in-one; Model B: Focused; Model C: Single biomarker.

Table 21: Progression-Free Survival: Treatment  $\times$  Biomarker Interactions Across Model Types

| Interaction          | Model A          | Model B          | Model C          |
|----------------------|------------------|------------------|------------------|
| CRP $\times$ Rx      | 0.22 (0.04-1.59) | 0.19 (0.03-1.34) | 0.21 (0.05-1.31) |
| TLR $\times$ Rx      | 0.87 (0.22-3.22) | 0.75 (0.20-2.66) | 0.53 (0.14-1.86) |
| TMB/BRAF $\times$ Rx | 0.48 (0.12-1.93) | 0.45 (0.11-1.78) | 0.36 (0.10-1.25) |

*Note:* Model A: All-in-one; Model B: Focused; Model C: Single biomarker.

## 6.2 Forest Plots



**Interpretation:**  $HR < 1$  indicates that biomarker-positive patients derive greater benefit from experimental treatment (reduced hazard of death/progression). The dashed vertical line at  $HR = 1$  represents no differential treatment effect by biomarker status.

## 7 Sensitivity Analysis: Ridge Regression

Ridge regression ( $\alpha = 0$ ) shrinks coefficients toward zero but does not eliminate any, providing a sensitivity analysis for the stability of our interaction estimates.

Table 22: Overall Survival: Firth-Corrected vs Ridge Regression Across Model Types

| Interaction   | Model A (All-in-One) |                 | Model B (Focused) |                 | Model C (Single) |                 |
|---------------|----------------------|-----------------|-------------------|-----------------|------------------|-----------------|
|               | Firth                | Ridge           | Firth             | Ridge           | Firth            | Ridge           |
| CRP x Rx      | 0.71 (0.17-3.56)     | 0.61 (SE: 2.20) | 0.97 (0.26-4.54)  | 0.53 (SE: 2.19) | 0.72 (0.20-3.23) | 0.46 (SE: 2.19) |
| TLR x Rx      | 2.63 (0.75-8.74)     | 0.20 (SE: 2.12) | 2.47 (0.75-7.75)  | 0.21 (SE: 2.11) | 1.98 (0.61-6.06) | 0.20 (SE: 2.11) |
| TMB/BRAF x Rx | 0.92 (0.29-2.91)     | 0.90 (SE: 2.04) | 0.89 (0.28-2.79)  | 0.91 (SE: 2.04) | 0.59 (0.20-1.72) | 1.00 (SE: 2.04) |

*Note:* Firth columns show HR (95% CI); Ridge columns show point estimates only.

Table 23: Progression-Free Survival: Firth-Corrected vs Ridge Regression Across Model Types

| Interaction   | Model A (All-in-One) |                 | Model B (Focused) |                 | Model C (Single) |                 |
|---------------|----------------------|-----------------|-------------------|-----------------|------------------|-----------------|
|               | Firth                | Ridge           | Firth             | Ridge           | Firth            | Ridge           |
| CRP x Rx      | 0.22 (0.04-1.59)     | 1.52 (SE: 2.26) | 0.19 (0.03-1.34)  | 1.32 (SE: 2.25) | 0.21 (0.05-1.31) | 0.92 (SE: 2.23) |
| TLR x Rx      | 0.87 (0.22-3.22)     | 0.14 (SE: 2.14) | 0.75 (0.20-2.66)  | 0.17 (SE: 2.13) | 0.53 (0.14-1.86) | 0.23 (SE: 2.12) |
| TMB/BRAF x Rx | 0.48 (0.12-1.93)     | 0.77 (SE: 2.06) | 0.45 (0.11-1.78)  | 0.68 (SE: 2.05) | 0.36 (0.10-1.25) | 0.74 (SE: 2.05) |

*Note:* Firth columns show HR (95% CI); Ridge columns show point estimates only.

**Interpretation:** Ridge regression applies an L2 penalty that shrinks all coefficients toward zero (HR toward 1.0), providing a sensitivity analysis for estimate stability across three model structures:

- **Model A (All-in-One):** Ridge regularizes all three biomarker main effects simultaneously while estimating all three interactions. This tests whether interaction estimates are stable when accounting for multicollinearity among biomarkers.
- **Model B (Focused):** Ridge regularizes all biomarker main effects but only one interaction is estimated per model. Comparing Model B Ridge to Model A Ridge reveals whether regularization effects differ when fewer interaction terms compete for variance.
- **Model C (Single-Biomarker):** Ridge regularizes only one biomarker term per model. Since there is minimal multicollinearity to address, Model C Ridge estimates should be closest to their Firth counterparts. Substantial shrinkage here would indicate instability in the biomarker-treatment relationship itself, not confounding.

**Key patterns to examine:** (1) Consistent shrinkage across all three models suggests the interaction estimate is inherently unstable; (2) Shrinkage only in Model A suggests multicollinearity with other biomarkers; (3) Similar Firth and Ridge estimates across all models supports robustness.

## 8 Summary

This analysis evaluated treatment effect heterogeneity across three candidate predictive biomarkers (CRP, TLR, TMB/BRAF) using Firth-corrected Cox regression with profile likelihood confidence intervals and penalized likelihood ratio tests.

### Key Findings:

1. **Model Robustness:** Treatment-biomarker interaction estimates were compared across three model specifications (all-in-one, focused, single-biomarker) to assess sensitivity to model structure.
2. **Multiple Testing:** Benjamini-Hochberg correction was applied to control the false discovery rate across the 6 primary interaction tests (3 biomarkers  $\times$  2 outcomes).
3. **Proportional Hazards:** The PH assumption was assessed using Schoenfeld residual tests and log-log survival plots. For overall survival, no significant PH violations were detected. For progression-free survival, significant violations were observed for TMB/BRAF main effect ( $p = 0.004$ ) and TMB/BRAF  $\times$  Rx interaction ( $p = 0.009$ ), with the global test also significant ( $p = 0.004$ ). These findings suggest time-varying effects for TMB/BRAF-related terms in PFS models, warranting caution in interpretation. The log-log plots show some crossing of survival curves for the treatment comparison, consistent with potential non-proportional hazards.
4. **Sensitivity Analysis (Ridge Regression):** Ridge regression was applied to all three model structures (A, B, C) to assess estimate stability. This comprehensive comparison reveals whether shrinkage patterns are consistent across model specifications or driven by specific modeling choices:
  - Consistent shrinkage across all three models suggests inherent instability in the interaction estimate
  - Shrinkage only in Model A (all-in-one) suggests multicollinearity with other biomarkers
  - Model C (single-biomarker) provides the most direct biomarker-treatment estimate with minimal confounding

Both methods answer different questions: Firth provides the best point estimate given the data structure, while Ridge prioritizes predictive stability. Comparing patterns across models helps distinguish true effect instability from model-specific artifacts.

**Methodological Note:** The use of Firth-corrected Cox regression addresses potential monotone likelihood issues and small sample bias. Profile likelihood-based confidence intervals and penalized likelihood ratio tests are more appropriate than Wald-based methods in this setting.

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